



Instructor: Date:

Learner: Regional Branch:

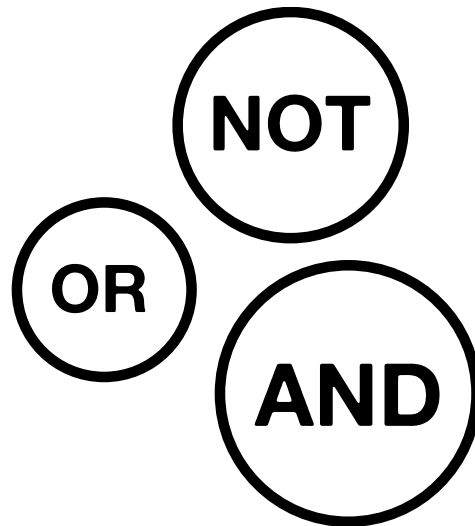
Program: Center:

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**Let's Start!**

Understanding Boolean Operators

**Learning Outcome:**

Describe the development of software programming in simple spoken and written situations.

Ask your partner:

- What Boolean operators do you know?
- Do you think Boolean operators are written differently depending on the programming language that you are using?

Can Do Statements:

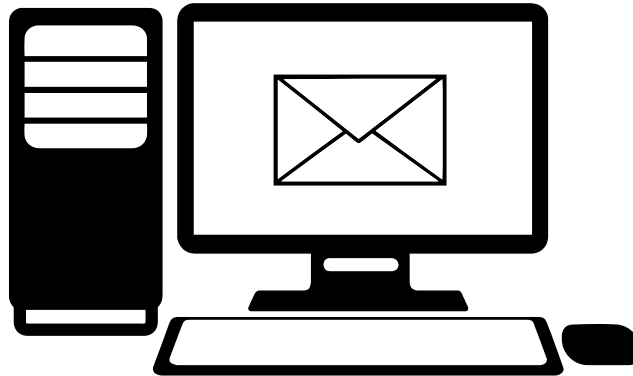
- I can talk about the combination of values for the AND operator.
- I can talk about the combination of values for the OR operator.
- I can discuss how to combine Boolean and comparison operators.





Let's Explore!

Robin sends Katy an email with more information about Boolean operators.



Instructions: Read the text below.

Hi, Katy!

How are you doing with the course on programming? Are you still studying the Boolean logic? I have more information that can be useful for you.

First, you need to remember that the operands are the statements that can be proven True or False, and that there are three Boolean operators: AND, OR, and NOT. Now, you can have different results if you combine True or False values depending on the operator that you use. Look at the following examples:

True AND True evaluates to True

True AND False evaluates to False

False AND True evaluates to False

False AND False evaluates to False

As you can see, the combination of two True values results in True, but all the other combinations result in False. Now, are we getting similar results when we use the OR operator? Look at the following combinations:

True OR True evaluates to True

True OR False evaluates to True

False OR True evaluates to True

False OR False evaluates to False

In the case of using OR as the operator, it is necessary that both values are False to evaluate to False.

It is common that Boolean expressions have comparison operators such as equality, greater than, less than, greater than or equal to, less than or equal to, and inequality.

For example, consider this question: "Are there more than 20 people studying in the library, and are there more than 20 computers?" We can express the questions like this: "Number_of_People > 20 AND Number_of_Computers > 20" Now, let's imagine a library with more than 20 people and computers. We can say that both values evaluate as true. When we substitute those values, we have True AND True. When you check the combinations of the AND operator, this entire expression evaluates as TRUE.

Hope this helps! Bye





Let's Practice!

Are you still taking programming lessons?

Instructions: Use the words from the box to correctly complete the sentences.

Word Bank

studying the Boolean logic / Is Katy reviewing the True / How are you doing / noticing that the combination of two / Is Anna saying

1. _____ with the course on programming?
2. Are you still _____?
3. _____ that operands are the statements that can be proven True or False?
4. _____ or False values depending on the operator?
5. Are they _____ True values results in True?



Let's Practice!

There are some comparison operators you might need to know!

Instructions: Select the correct answer.

1. Boolean expressions _____ comparison operators such as equality, greater than, less than, greater than or equal to, less than or equal to, and inequality.
 - a. can take
 - b. can have
 - c. can speak
 - d. can break
2. Comparisons operators _____ to determine if a statement is True or False.
 - a. can be
 - b. can see
 - c. can help
 - d. can take
3. We _____ equality with the symbol "=".
 - a. can represent
 - b. can see
 - c. can be
 - d. can make





Extension Activity

How does Boolean Logic work?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Do some further research about Boolean Logic.
- Take notes about key ideas.
- Create a brochure to explain how Boolean Logic works and why it is important for computer science.
- Use simple sentences.
- Share your brochure with a partner.

Self-Assessment

Yes No

- I deepen my knowledge about Boolean Logic. ☐ ☐
- I take notes of main ideas. ☐ ☐
- I express information through a brochure. ☐ ☐
- I exchange information with partners. ☐ ☐



Target Words

- both** (pronoun): referring to two things or people that are mentioned before
e.g. In the case of using OR as the operator, it is necessary that both values are False to evaluate to False.
- substitute** (verb): to replace something with something else
e.g. When we substitute those values, we have True AND True. If you check the combinations of the AND operator, this entire expression evaluates as TRUE.
- prove** (verb): to say that something is true or false after analysis or test
e.g. First, you need to remember that the operands are the statements that can be proven True or False, and that there are three Boolean operators: AND, OR, and NOT.
- comparison** (noun): an analysis of two or more statements to find similarities or differences
e.g. A warehouse or a group of machines are examples of impersonal cost centers.
- entire** (adjective): complete
e.g. When you check the combinations of the AND operator, this entire expression evaluates as TRUE.
- consider** (verb): to think about something
e.g. For example, consider this question: "Are there more than 20 people studying in the library, and are there more than 20 computers?"



Learning Points

Present Continuous Tense in Questions

You use the **present continuous** tense to talk about actions that are happening now.

To make questions in the **present continuous tense** you need to follow the following structure:

To make questions in the present continuous tense you need to follow this structure:

(Wh) + **be + subject + verb** in -ing + ?

Look at the following examples:

Are you taking the programming course?

What programming language **are you using**?

Are you using the correct Boolean operator?

Examples

- *How are you doing with the course on programming?*
- *Are you still studying the Boolean logic?*
- *Now, are we getting similar results if we use the OR operator?*



Learning Points

How to Express Operators in Different Languages

There are three Boolean operators: AND, OR, and NOT

Depending on the programming language that you are using, you need to consider how the Boolean operators are written.

Language	AND	OR	NOT
Mathematics	$\wedge$	$\vee$	$\neg$
"C" or C++	&&		!
SQL	AND	OR	NOT
Pascal	AND	OR	NOT
Perl	&		!
Java	&		!
Basic	AND	OR	NOT